

Professor's Questions: Full Answers

Experiment 8 Follow-Up + Experiments 9-12

Q1 — Which surahs are in each of the 8 clusters, and which 3 are outliers?

The 3 outliers: Surah 14 (Ibrahim), Surah 26 (Ash-Shu'araa), Surah 72 (Al-Jinn).

Cluster	Size	Surah numbers
Cluster 1	22	1, 56, 74, 77, 78, 79, 80, 81, 85, 86, 87, 88, 90, 91, 92, 94, 96, 100, 101, 102, 107, 114
Cluster 2	10	19, 20, 23, 36, 38, 43, 50, 71, 76, 110
Cluster 3	26	2, 4, 5, 6, 8, 9, 10, 11, 13, 24, 31, 33, 35, 42, 46, 48, 49, 57, 58, 59, 60, 62, 63, 64, 65, 66
Cluster 4	8	15, 44, 51, 52, 54, 68, 73, 97
Cluster 5	22	37, 53, 55, 69, 70, 75, 82, 83, 84, 89, 93, 95, 99, 103, 104, 105, 106, 108, 109, 111, 112, 113
Cluster 6	7	3, 12, 22, 28, 34, 41, 61
Cluster 7	6	21, 25, 27, 47, 67, 98
Cluster 8	10	7, 16, 17, 18, 29, 30, 32, 39, 40, 45
Outliers	3	14, 26, 72

(Cluster 0 = all 35 poets, no surahs — full surah names are in the notebook's exported CSV.)

Q2 — Which color represents the outliers?

Gray. Same convention used for outliers in every figure throughout this whole project.

Q3 — New graph with outliers clearly marked

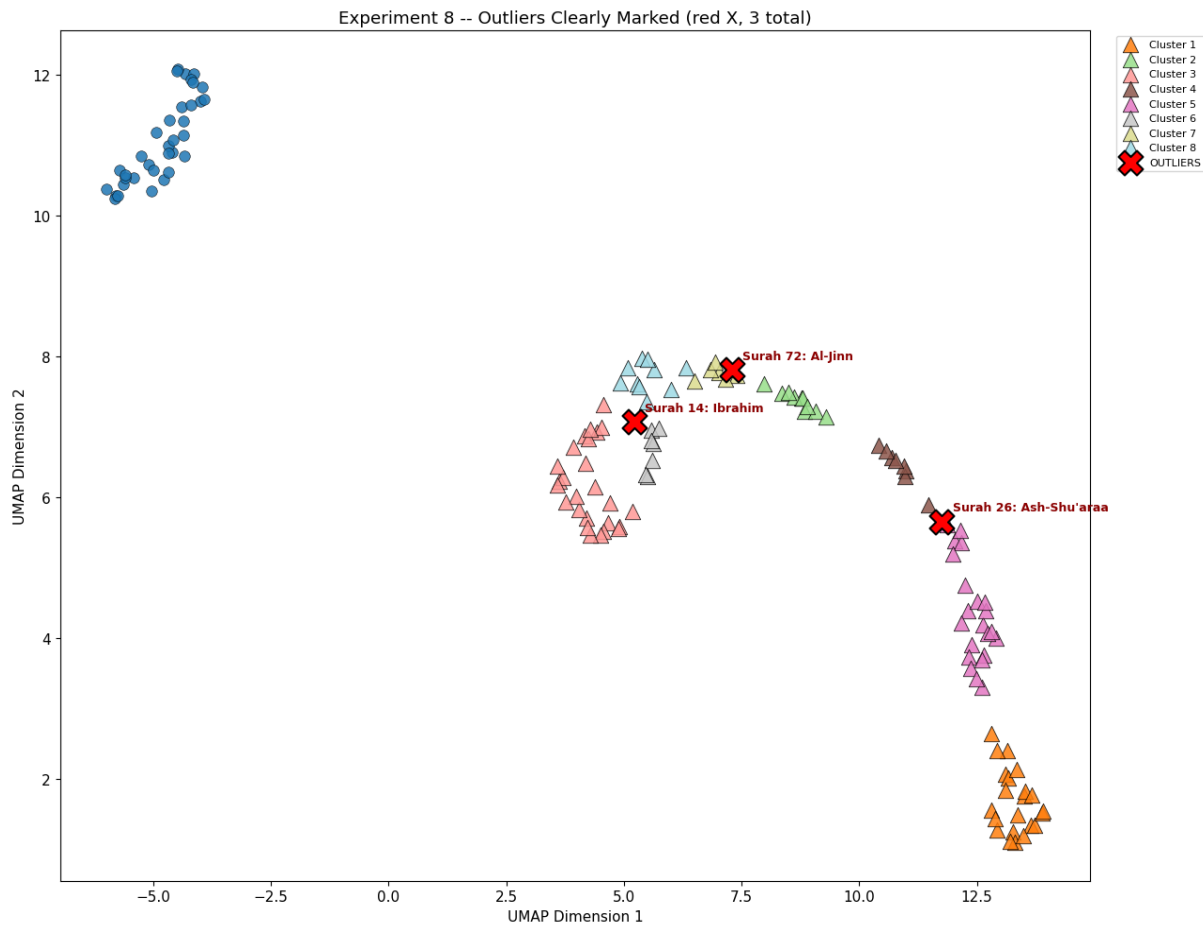


Figure 1. Each of the 3 outlier surahs marked with a large red X and labeled by name.

Q4 — New graph with surah numbers labeled

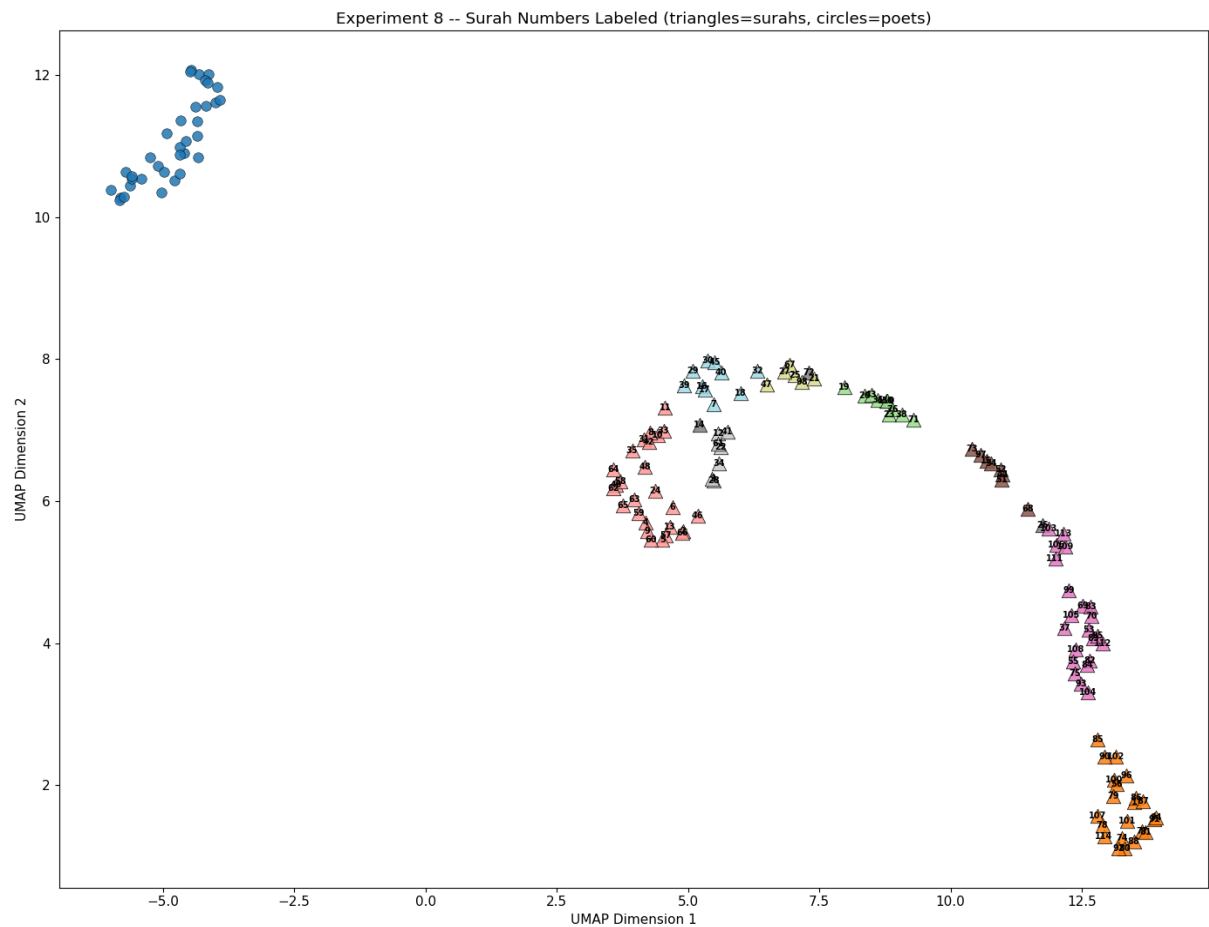


Figure 2. Every surah triangle labeled with its number.

Experiment 9 — 60 Hizb + 35 Poets

Metric	Exp 9 (60 hizb)	Exp 10 (30 hizb-pairs)
Total entities	95	65
Clusters found	4	2
Outliers	0	0
Silhouette score	0.739	0.892 (very clean)
Units sharing a cluster with poets	4 / 60 (6.7%)	5 / 30 (16.7%)

Both finer (hizb) and coarser (hizb-pair) groupings show the same pattern as Experiment 8: the large majority of Quranic units separate cleanly from poets. Coarser grouping (fewer, bigger units) shows somewhat more mixing (16.7% vs 6.7%) and dramatically higher silhouette — fewer, more distinct units are easier for the algorithm to separate cleanly overall.

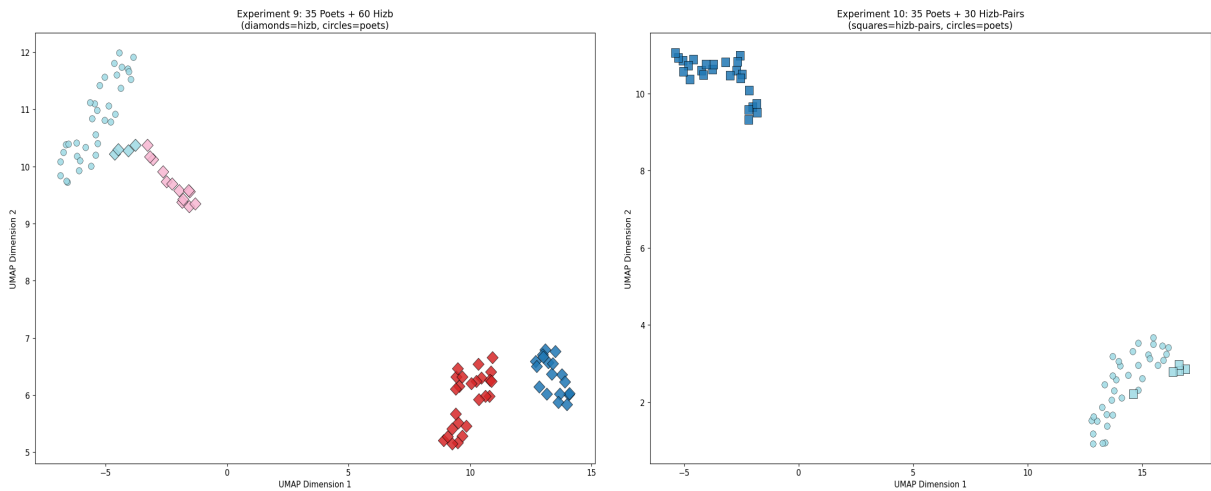


Figure 3-4. Experiment 9 (left) and 10 (right). Poets and Quranic units separate into distinct regions in both.

Experiment 11 & 12 — Random Sampling (10 trials each)

	Exp 11 (1 random surah)	Exp 12 (10-surah bundle)
Outlier	1 / 10 trials	0 / 10 trials
Shared a cluster with poets	9 / 10 trials	10 / 10 trials

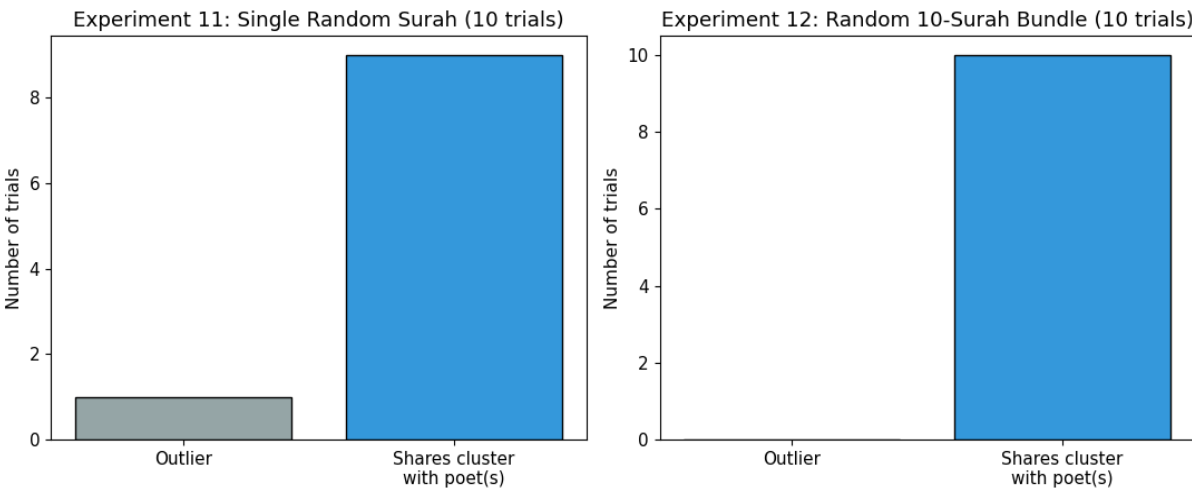


Figure 5. Outcome across all 10 trials, both experiments.

This looks like it contradicts Experiments 8-10 — it doesn't, and the reason why matters more than the numbers themselves.

In Experiments 8, 9, and 10, MANY Quranic units (114 surahs, 60 hizb, 30 pairs) were present together, so they had enough of each other to form their OWN separate clusters. In Experiments 11 and 12, only ONE Quranic unit is present at a time, surrounded by 35 poets. The clustering method (HDBSCAN) requires several similar points close together to form a new cluster — a single lone point can never do that by itself, no matter how distinct its actual style is. So a lone unit has only two possible outcomes: join whichever existing poet cluster happens to be nearest, or get marked an outlier. With 35 poets already forming several clusters and only 1 foreign point, joining a poet cluster is close to the default outcome — not necessarily evidence that the single surah's style genuinely resembles poetry.

What would settle this properly: a control test — inserting a single clearly-unrelated text (a modern news article, or random text) the same way, to see if it ALSO gets pulled into a poet cluster by default. If it does, that confirms Experiments 11-12 are mostly measuring clustering mechanics, not style similarity. This wasn't tested here and would be a strong next step.